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High-capacity totes collapser

Abstract

A tote collapser can be used for unlatching and collapsing walls of a shippable tote. The collapser includes a mounting frame, and a set of collapser brackets. The mounting frame can include a first frame member and a second frame member movable relative to each other. The set of collapser brackets can include a first and a second collapser brackets attached to the first frame member, and a third and a fourth collapser brackets attached to the second frame member. Each collapser bracket can include an indexer shaped to align the tote in a specified orientation, a retainer extending perpendicularly from the indexer and configured to engage with top side of the sidewalls, a bumper shaped to engage with a latch on a sidewall of the tote, and a receiving space between the indexer and the bumper to receive a thickness portion of the sidewall of the tote.

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Background/Summary

BACKGROUND

(1) Inventory systems, such as those in warehouses, supply chain distribution centers, airport luggage systems, and custom-order manufacturing facilities, face significant challenges in storing inventory items. Typically, high-capacity totes are used to store and transport items through these inventory systems. Once the items in the totes are stored at an inventory storage location, moving or shipping empty totes to a secondary location may be prohibitively expensive, limiting the ability of the location to accommodate additional items. The user of high-capacity totes, such as those that have large volumes and straight sidewalls, may be capable of accommodating additional items. However, because the stacking density of these high-capacity totes is lower, use of them may result in inefficient use of space within the inventory systems.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various embodiments in accordance with the present disclosure will be described with reference to the drawings, in which:

(2) FIG. **1** illustrates an example station of an inventory system employing a tote collapser with an automatic linear actuator, according to various embodiments;

(3) FIG. **2** illustrates an example tote collapser with a tote in un-collapsed state, according to various embodiments;

(4) FIG. **3** illustrates the tote collapser of FIG. **2** with the tote in a collapsed state, according to various embodiments;

(5) FIG. **4** illustrates an example tote, according to various embodiments;

(6) FIG. **5** illustrates an example tote collapser bracket of FIG. **2** aligned with a tote in an un-collapsed state, according to various embodiments;

(7) FIG. **6** illustrates the tote collapser bracket of FIG. **2** engaged with the tote in an un-collapsed state, according to some embodiments;

(8) FIG. **7** illustrates a perspective view of an example tote collapser bracket without a retainer;
(9) FIG. **8** illustrates a perspective view of another example of a tote collapser bracket;
(10) FIG. **9** illustrates a frame member coupled with the tote collapser brackets of FIG. **7**;
(11) FIG. **10** illustrates a manually operated tote collapser system with the frame member of FIG. **9**;
(12) FIG. **11** illustrates a tote collapsing operation using the manually operated tote collapser system of FIG. **10**;
(13) FIG. **12** illustrates a collapsed tote with the manually operated tote collapser system of FIG. **10**; and
(14) FIG. **13** illustrates a tote collapser with a manually operated linear actuator, according to some embodiments.

DETAILED DESCRIPTION

(15) In the following description, various embodiments will be described. For purposes of explanation, specific configurations and details are set forth in order to provide a thorough understanding of the embodiments. However, it will also be apparent to one skilled in the art that the embodiments may be practiced without the specific details. Furthermore, well-known features may be omitted or simplified in order not to obscure the embodiment being described.

(16) Various embodiments herein are directed to features that may be utilized for high-capacity totes of inventory systems. The present disclosure provides totes and a tote collapser to advantageously maximize a cubic volume of storage capacity of a tote and to maximize a density of stacked totes. The sidewalls of the totes herein can be collapsed to facilitate nesting of totes. The tote collapser herein also facilitates efficient collapsing of high-capacity totes leading to increased efficiencies in terms of time, and space, while making tote collapsing an ergonomically friendly operation.

(17) An example tote collapser may include features for unlatching and collapsing sidewalls of a collapsible tote such as a shippable tote. In a collapsible tote, one sidewall can be latched to another and can be foldable or collapsible relative to a bottom wall. The sidewalls can be hingedly coupled to the bottom wall. With the tote collapser, sidewalls of an empty tote can be collapsed, with minimal human intervention or effort, to a compact size for transporting.

(18) In various embodiments, the tote collapser can include a mounting frame and a set of collapser brackets coupled to frame members of the mounting frame. In one example, the frame members can be moved via a linear actuator. The linear actuator can be moved automatically or manually operated. In another example, the tote collapser can be mounted to a table and used by an operator to manually place and collapse the tote without the linear actuator.

(19) In various embodiments, each collapser bracket of the tote collapser can include an indexer shaped to align the tote in a specified orientation, a bumper shaped to engage with a latch on a sidewall of the tote, and a receiving space between the indexer and the bumper to receive a portion of the sidewall of the tote. Optionally, the collapser bracket can include a retainer extending perpendicularly from the indexer and configured to engage with top side of the sidewalls for limiting upward vertical displacement of the tote sidewall during use.

(20) In operation, the collapser brackets can be moved towards respective latches of the sidewalls of the tote. The indexer can align sides of the tote parallel to a frame member. The bumper can push the latch on the sidewalls of the tote to a released state and further may push the released sidewalls toward a collapsed state. During an initial move, the contoured surface portion of the bumper engages and push a latch as a collapser bracket is advanced relative to the tote. Upon moving further, the flat surface of the bumper can push a face of the sidewall and collapse it over a bottom wall. This way, the tote can be collapsed or folded with the sidewalls folded over each other. Several folded totes can be stacked on top of each other to enable transportation.

(21) Referring now to the drawings, in which similar identifiers refer to similar elements, FIG. **1** illustrates an example tote collapser station **10** of a fulfillment center according to various

embodiments. In the illustrated embodiment of FIG. **1**, the station **10** can include a conveyor **12**, and a tote collapser **20**. The tote collapser **20** may be located at an end or portion of the conveyor **12**, in line with the conveyor **12**, and/or otherwise aligned to be positioned for acting on items supplied by the conveyor **12**. The conveyor **12** can receive a tote **100** at one portion and move the tote **100** to the tote collapser **20** at another portion. In the illustrated embodiment, the tote collapser **20** can be configured to automatically collapse sidewalls of the tote **100**. For example, the tote collapser **20** can include a mounting frame **200**, an automatic linear actuator **210**, and a set of collapser brackets **300**. The automatic linear actuator **210** can be configured to move the set of collapser brackets **300** relative to the tote **100**. For example, the automatic linear actuator **210** can be a piston cylinder arrangement coupled to frame members of the mounting frame **200**. Movement of the piston within the cylinder can be automatically controlled to control movement of the collapser brackets **300**. The automatic linear actuator **210** can be controlled remotely, e.g., using a controller or other computer systems of the fulfillment center. While a linear actuator **210** is shown and described with respect to FIG. **1**, the present disclosure is not limited to use of a linear actuator. Any other suitable actuator or other mechanism capable of performing the function described herein may be employed to move the frame members. For example, other suitable actuators may include a lead screw, a solenoid, a motor operated actuator, or other mechanical or electromechanical actuators.

(22) FIGS. **2** and **3** illustrate an example of the tote collapser **20** and a tote **100** in un-collapsed state and in a collapsed state, respectively. In many embodiments, the tote collapser **20** can include a mounting frame **200** and a set of collapser brackets **300**. The tote **100** can be received at the mounting frame **200** and aligned with the set of collapser brackets **300**. As shown in FIG. **2**, the tote **100** is received in an un-collapsed state, where all sidewalls (e.g., **101**, **102**, **103**, **104**) of the tote **100** are in upright state and coupled to each other. As shown in FIG. **3**, the set of collapser brackets **300** are moved towards an opposing couple of sidewalls (e.g., **102** and **104**) of the tote **100** to unlatch and collapse the sidewalls (e.g., **101**, **102**, **103**, **104**). This way, the tote **100** can be in the collapsed state. In the collapsed state, the sidewalls (e.g., **101**, **102**, **103**, **104**) of the tote **100** can fold on top of each other or lie in a plane to form a compact platform. Several such collapsed totes can be stacked on top of each other thereby saving space during shipping/transporting of empty totes.

(23) In the illustrated embodiment, the mounting frame **200** can include a first frame member **201** and a second frame member **202** movable relative to each other. For example, the linear actuator **210** can be mounted on top of the mounting frame **200** and coupled to the frame members **201** and **202** to move them towards or away from each other. The linear actuator **210** may be an automatically operated piston cylinder, where a piston **211** can be attached to the first frame member **201** and an end of a cylinder **212** can be attached to the second frame member **202**. When the piston **211** moves in and out of the cylinder **212**, the frame members **201**, **202** can move towards or away from each other. The linear actuator **210** can be located approximately at the center of the frame members **201** and **202**. This allows the frame members **201** and **202** to move relatively parallel to each other and be parallelly aligned a length of the sidewall (e.g., a front sidewall and a back sidewall) of the tote **100**. Such alignment can facilitate simultaneous engagement of each of the set of collapser brackets **300** with the tote **100**. In some cases, the sidewalls of the tote **100** may flex in response to a force. Hence, proper alignment of the collapser brackets **300** with the tote **100** can prevent any misorientation of the tote **100** during the collapsing process. The present disclosure provides several features that can be used for proper alignment of the tote **100** with respect to the collapser brackets **300**.

(24) In the illustrated embodiment, the set of collapser brackets **300** can include four collapser brackets **300** corresponding to corners of the tote **100**. For example, a first and a second collapser brackets **300** can be coupled to the first frame member **201** and spaced from each other. Similarly, a third and the fourth collapser brackets **300** can be coupled to the second frame member **202** and

spaced from each other. A lateral distance (e.g., along x-direction) between the collapser brackets **300** on the first frame member **201** and those on the second frame member **202** can each be the same. For example, the lateral distance can correspond to a distance L**1** between latches on the sidewalls of the tote **100** (e.g., see distance L**1** between latches **110** in FIG. **4**).
(25) In the illustrated embodiment, e.g., in FIG. **2**, the collapser bracket **300** can include a base **301**, an indexer **310**, a bumper **320**, a receiving space **330** and optionally a retainer **340**. The base **301** can be configured to be attached to a frame member (e.g., **201**, **202**) or other mounting surfaces. For example, the base **301** can include holes to receive a fastener. The indexer **310** and the bumper **320** can extend perpendicularly (e.g., along y-axis) to the base **301**. The indexer **310** can be longer in length than the bumper **320**. This may allow the indexer **310** to engage with the tote sidewalls and align the sidewalls with the frame members **201**, **202** prior to engaging the bumper **320**. The receiving space **330** can be a space between the indexer **310** and the bumper **320**. The receiving space **330** can be sized to receive a thickness portion (e.g., t**1** of sidewall **103** shown FIGS. **5** and **6**) of the sidewall of the tote.
(26) The indexer **310** can be or include an indexer projection extending or projecting from the base **301**. In many embodiments, the indexer **310** can include a shaped portion **311** (e.g., an inclined portion) to align the tote **100** in a specified orientation. For example, the indexer **310** can facilitate left-right centering of the tote **100** with respect to the mounting frame. In the illustrated embodiment, the indexer **310** can include an inclined surface **311** that is angled away from the receiving space **330** and the bumper **320**. The inclined surface **311** can extend relatively beyond a length of the bumper **320**. For example, the indexer **310** can have an indexer length (e.g., see L**2** in FIG. **5**) greater than a bumper length (e.g., see L**3** in FIG. **5**) of the bumper **320**. The inclined surface **311** extends distally from a base portion (e.g., see L**4** in FIG. **5**) attached to the base **301**.
(27) The inclined surface **311** can allow a sidewall of the tote to be aligned within the receiving space **330**. If the tote **100** is not properly aligned or oriented with respect to the set of collapser brackets **300**, the sidewalls of the tote **100** can slide along the inclined surface **311** and reorient the tote **100**. For example, if the tote **100** is not centered with respect to the mounting frame, or slightly angularly misoriented, the inclined surface **311** can center and/or reorient the tote **100** so that the face of the sidewalls **102** and **104** (e.g., FIG. **2**) are substantially parallel to the frame members **201**, **202**. This way, the indexer **310** can properly align the tote **100** before the bumper **320** is engaged with the tote **100** to unlatch the sidewalls.
(28) Additionally or alternatively, the retainer **340** can be or include a retainer projection extending perpendicularly (e.g., along z-direction) from the indexer **310** and configured to engage with top side of the sidewalls (e.g., see FIG. **3**). As may be best seen in FIG. **2**, the retainer **340** can be an L-shaped element including a leg **341** and a flange **342**. The leg **341** can extend perpendicularly (e.g., along z-direction) from the indexer **310**, and the flange **342** can be perpendicular (e.g., along x-direction) to the leg **341**. In the illustrated embodiment, the retainer **340** may be disposed adjacent to the receiving space **330** such that the flange **342** extends over or above the receiving space **330**. When a portion of the sidewall (e.g., **101**, **103**) is received within the receiving space **330**, the flange **342** can engage with the topside of the sidewall and prevent the tote from misaligning or misorienting during a tote collapsing process.
(29) In many embodiments, the bumper **320** can be or include a bumper projection shaped to engage with a latch (e.g., **110**) on a sidewall of the tote (e.g., **100**). The bumper **320** can have a contoured surface **321** and a flat surface **322**. The contoured surface **321** can be configured to progressively push the latch (e.g., **110**) of the tote (e.g., **100**) and unlatch the sidewall (e.g., in a camming action as the bumper **320** is pressed inward against the latch (e.g., **110**) of the sidewall). The contoured surface **321** can be a curved surface contoured to push the latch **110** (e.g., push a tab **111** (e.g., FIG. **3**) of the latch **110**). As another example, the contoured surface **321** can be an inclined surface, curved, and/or other suitable shapes. The contoured surface **321** may bound a portion of the receiving space **330** and/or curve away from the receiving space **330**. The flat surface

322 of the bumper **320** can extend (e.g., distally from the base **301**) from the contoured surface **321**. The flat surface **322** can be configured to engage with a face of the sidewall (e.g., a front face of the sidewall **102**) and collapse the sidewall (e.g., **102**) of the tote **100** (e.g., when the latch **110** has been released by the bumper **320**).

(30) Referring to FIGS. **2-4**, the tote **100** can include four sidewalls (e.g., a right sidewall **101**, a front sidewall **102**, a left sidewall **103**, and a rear sidewall **104**), a bottom wall (e.g., **105**) and an open top to receive or remove items. The tote **100** can have a length L (see FIG. **4**), and width W (see FIG. **2**). One sidewall can be coupled to another sidewall using a latch **110** (e.g., see enlarged portions A**1** and A**2** in FIGS. **2** and **3**). The latch **110** can be located on one of the sidewalls. In the illustrated embodiments, two latches **110** can be located on the left sidewall **103** and two latches **110** can be located on the right sidewall **101**. For example, as shown in an enlarged left portion A**1** of the tote **100** in FIG. **2**, the front sidewall **102** is coupled to the left sidewall **103** via a latch **110**. Similarly, the front sidewall **102** can be coupled to the right sidewall **101** via another latch **110**, the back sidewall **104** can be coupled to the right sidewall **101** via yet another latch **110**, and the back sidewall **104** can be coupled to the left sidewall **103** via yet another latch **110**. Referring to FIG. **4**, the front sidewall **102** is coupled with the left sidewall **103** and the right sidewall **101** via respective latches **110**.

(31) Referring to an enlarged left sidewall portion A**2** in FIG. **3**, the latch **110** can include a tab **111**, a lip **112**, and a pivot arm **113**. The latch **110** can be formed as a cantilever on a sidewall, can be spring loaded, or can have other lock/latch configurations. For example, as shown in the enlarged portion A**2**, the latch **110** is cantilevered such that when a force is applied at the tab **111**, the latch **110** moves away (e.g., in direction D**1** shown in FIG. **4**) about the pivot arm **113**. In many embodiments, such force is applied by the contoured surface (e.g., **321**) of the bumper (e.g., **320**) of the collapser bracket (e.g., **300**). The lip **112** (e.g., on the sidewall **103**) can engage with an inside surface of an adjacent sidewall (e.g., **102**) to lock the sidewalls together. When the tab **111** is moved sideways (e.g., along direction D**1**), the lip **112** can be disengaged from the adjacent sidewall (e.g., **102**) so that the sidewalls can be collapsed or folded over one another. It can be understood that the present disclosure is not limited to a particular latch, and other appropriate latches (e.g., having a plate-like form, a cylindrical, or other shapes) may be used to latch the sidewalls of the tote **100**.

(32) In operation, when the linear actuator **210** (e.g., in FIG. **1**) is actuated the frame members **201**, **202** moves towards each other to cause the set of collapser brackets **300** to undergo motion toward the tote **100**. In response to the motion, the indexer **310** aligns the tote relative to the mounting frame **200** and guides the thickness portion (e.g., t**1**) of the sidewall (e.g., **103**) of the tote **100** toward the receiving space. Also, in response to the motion, the bumper **320** engages with the latch **110** and unlatches the latch **110** on the sidewall (e.g., **103**) of the tote **100** to facilitate collapse of the sidewalls of the tote **100**. The retainer **340** keeps the sidewall from moving upward.

(33) FIGS. **5** and **6** illustrate collapsing operation of the tote using the collapser bracket **300**. In the illustrated embodiment, the retainer (e.g., **340**) is omitted to better illustrate the indexing and the unlatching process achieved by the collapser bracket **300**. Also, portions of only two sidewalls **102** and **103** are shown for illustration purposes to explain the collapsing operation, without limiting the scope of the disclosure. Referring to FIGS. **2** and **5**, the tote **100** is appropriately positioned via the inclined surface **311** of the indexer **310**. For example, the tote **100** is positioned such that a thickness portion t**1** of the left sidewall **103** of the tote **100** is in front of the receiving space **330**, and the adjacent sidewall e.g., the front sidewall **102** faces the bumper **320**. The tote **100** may travel along the inclined surface **311** of the indexer **310** to reach a suitably aligned position, for example. The latch **110** on the left sidewall **103** is in latched position so that the sidewalls **102** and **103** are locked in place. For example, the tab **111** of the latch **110** is in an extended state. In order to unlatch the latch **110**, the collapser bracket **300** is moved closer toward the tote **100** to engage the bumper **320** with the tab **111** of the latch **110**.

(34) Referring to FIGS. **3** and **6**, as the collapser bracket **300** moves relative to the tote **100**, the contoured surface **321** of the bumper **320** engages with the tab **111** of the latch **110**. As the contoured surface **321** contacts the tab **111**, the contoured surface **321** starts pushing the tab **111** towards the left sidewall **103** and unlatches the front sidewall **102** from the left sidewall **103**. Upon moving further, the flat surface **322** of the bumper **320** pushes the unlatched front sidewall **102**, causing the now-released front sidewall **102** to collapse (see FIG. **3**) over the bottom wall **105**. The left sidewall **103** can be retained within the receiving space **330** in an upright position. Similarly, as shown in FIG. **3**, the back sidewall **104** is collapsed over the bottom wall **105**. Additionally, the remaining sidewalls **101** and **103** can be collapsed over folded sidewalls **102** and **104**, thereby flattening the tote **100**. In some embodiments, the sidewalls **101-104** can be hingedly coupled to the bottom wall **105**. This way, the sidewalls **101-104** can be folded over the bottom wall **105** when the latches **110** are unlatched by the respective collapser brackets **300**. One or more flattened totes **100** can be stacked on top of each other to save space and easily transport several totes **100**.
(35) The present disclosure is not limited to the collapser bracket **300** or the mounting frame **200** configuration. Other collapser bracket configuration and mounting methods are possible. For example, FIGS. **7**, **8** and **11** illustrate other examples of the collapser brackets and FIGS. **9**, **10** and **13** illustrate other examples of mounting the collapser brackets.
(36) FIG. **7** illustrates a perspective view of an example tote collapser bracket **500** without a retainer. The collapser bracket **500** is substantially similar to the collapser bracket **300** in structure and function, except for the retainer (e.g., **340**). For example, the collapser bracket **500** can include a base **501**, an indexer **510**, a bumper **520** and a receiving space **530**. The indexer **510** can include an inclined surface **511**. The bumper **520** can include a contoured surface **521** and a flat surface **522**. The collapser bracket **500** can be an integrally formed solid piece. The collapser bracket **500** can be made of plastic or metal. The collapser bracket **500** can be manufactured using an additive manufacturing process (e.g., **3**D printer). In some embodiments, the bumper **520** may be solid, while the bumper **320** may be hollow. Although, both the indexer **310** and **510** are shown as hollow, they could be solid.
(37) FIG. **8** illustrates a perspective view of another example of a tote collapser bracket **600**. The collapser bracket **600** can be substantially similar to the collapser bracket **300** in structure and function, except for the retainer (e.g., **340**). For example, the collapser bracket **600** can include an indexer **610**, a bumper **620** and a receiving space **630**. The indexer **610** can include an inclined surface **611**. The bumper **620** can include a contoured surface **621** and a flat surface **622**. The collapser bracket **600** can be made of two thin plates **603** and **604** parallel to each other. The indexer **610** and the bumper **620** may be stamped out of or machined from a metal plate. The space between the plates **603** and **604** maybe hollow. In the illustrated embodiment, the space between the plates **603** and **604** can include a structural support column so that the plates do not collapse onto each other.
(38) FIG. **9** illustrates a frame member **900** coupled with two tote collapser brackets **500**. The tote collapser brackets **500** can be spaced from each other. For example, a distance between two bumpers **520** of the respective collapser brackets **500** can be L**1** (e.g., see FIG. **4**). The frame member can be fixed to a wall or a support surface. Accordingly, different collapser systems can be configured. For example, two frame **900** can be mounted on a mounting frame and linearly actuated (e.g., as shown in FIGS. **1** and **13**). In another example, a frame **900** can be fixed to a table to collapse a back side of a tote (e.g., **100**) using the collapser brackets **500** and to implement manual collapsing at a front side of the tote (e.g., **100**).
(39) FIGS. **10-12** illustrate a manually operated tote collapser system with the frame member **900** coupled to a table frame of a table **1000** or a wall. As shown in FIG. **10**, the table **1000** can include a receiving platform **1001** on which the tote **100** can be placed and pushed against the frame **900** to collapse the tote **100**. Referring to FIG. **11**, a tote **100** can be received on the receiving platform **1001**. An operator **1105** may align and push the tote **100** against the frame **900** so that the collapser

brackets **500** can unlatch the back sidewalls of the tote **100**. Additionally, the operator **1105** may wear hand gloves **1110** with another set of collapser brackets **700** attached to the hand gloves **1110**.
(40) The glove mountable collapser bracket **700** can be substantially similar to the collapser bracket **500** in structure and function. For example, the glove mountable collapser bracket **700** can include an indexer **710**, a bumper **720** and a receiving space **730**. The indexer **710** can be a straight projection **711** without an inclined surface. The bumper **720** can include a contoured surface **721** and a flat surface **722**. The operator **1105** can position the hand gloves **1110** such that the respective glove mountable collapser brackets **700** can be aligned with latches on the front sidewalls of the tote **100**. The operator **1105** can place the indexer **710** against the left and right sidewalls, as shown in FIG. **11**. The bumper **720** gets aligned with the latches (e.g., **110** in FIG. **4**). The operator **1105** can then push the hand gloves **1110** further causing the sidewalls to unlatch.
(41) Once the sidewalls are unlatched, the front and back sidewalls **102** and **104** collapse on top of each other, as shown in FIG. **12**. Further, the operator **1105** can fold down the left and right sidewalls on the other sidewalls. This way, the tote **100** can be transitioned to a collapsed state. In the collapsed state, the tote **100** is flattened, resulting in a compact and cubic shape. Several such collapsed totes **100** can be easily stacked and transported.
(42) In some embodiments, the collapser brackets **700** may be optional or omitted from the gloves. For example, the operator **1105** can manually push, e.g., using fingers or other parts of their hands, the latch **110** on the tote **100** to unlatch one side (e.g., **102**) of the tote **100**, while the other side (e.g., **104**) of the tote **100** may be collapsed via the collapser brackets **500** on the frame **900**, as shown in FIG. **10**.
(43) FIG. **13** illustrates a tote collapser with manually operated linear actuator. The tote collapser can include a mounting frame **1300** and a linear actuator **1310** mounted on top of the mounting frame **1300**. The mounting frame **1300** can include frame members **1301**, **1302** with a set of collapser brackets **500**. The frame members **1301**, **1302** can be examples of the frame **900** (see FIG. **9**). The frame members **1301**, **1302** can be coupled to the linear actuator **1310** to move the frame members **1301**, **1302**. The linear actuator **1310** can include a gear **1311**, a shaft **1312**, and a lever **1313**. As an example, the gear **1311** can be a rack and pinion. The lever **1313** can be coupled to the shaft **1312**. An operator can manually rotate the lever **1313** along a direction R**1** to rotate the pinion of the gear **1311** causing the rack to linearly move the frame members **1301**, **1302** to move towards or away from each other. As the frame members **1301**, **1302** move towards each other, the collapser brackets **500** unlatch the sidewalls of the tote **100**, e.g., as discussed with respect to FIGS. **2-6**.
(44) In the illustrated embodiments, a collapser bracket (e.g., **300**, **500**, **600**, and **700**) includes an indexer, a bumper, and a receiving space integrally formed on a base. However, in some embodiments, the indexer and the bumper may be separately formed. For example, as illustrated in FIG. **13**, indexers **1320** can be attached to an indexer frame **1321**. The indexers **1320** can be structured similar to the other indexers discussed herein. For example, the indexers **1320** can include an inclined surface and spaced from each other to allow left-right centering of a tote when engaged. The indexer frame **1321** may be movable e.g., via an actuator **1350** relative to the tote **100**. The actuator **1350** can be a linear actuator, a lead screw, or other suitable actuator.
(45) Based on the disclosure and teachings provided herein, a person of ordinary skill in the art will appreciate other ways and/or methods to implement the various embodiments. The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense. It will, however, be evident that various modifications and changes may be made thereunto without departing from the broader spirit and scope of the disclosure as set forth in the claims.
(46) Other variations are within the spirit of the present disclosure. Thus, while the disclosed techniques are susceptible to various modifications and alternative constructions, certain illustrated embodiments thereof are shown in the drawings and have been described above in detail. It should be understood, however, that there is no intention to limit the disclosure to the specific form or

forms disclosed, but on the contrary, the intention is to cover all modifications, alternative constructions, and equivalents falling within the spirit and scope of the disclosure, as defined in the appended claims.
(47) The use of the terms “a” and “an” and “the” and similar referents in the context of describing the disclosed embodiments (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The terms “comprising,” “having,” “including,” and “containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to,”) unless otherwise noted. The term “connected” is to be construed as partly or wholly contained within, attached to, or joined together, even if there is something intervening. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate embodiments of the disclosure and does not pose a limitation on the scope of the disclosure unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the disclosure.
(48) Disjunctive language such as the phrase “at least one of X, Y, or Z,” unless specifically stated otherwise, is intended to be understood within the context as used in general to present that an item, term, etc., may be either X, Y, or Z, or any combination thereof (e.g., X, Y, and/or Z). Thus, such disjunctive language is not generally intended to, and should not, imply that certain embodiments require at least one of X, at least one of Y, or at least one of Z to each be present.

Claims

1. A tote collapser system for collapsing sidewalls of a tote, comprising: a mounting frame comprising a first frame member and a second frame member movable relative to each other; an indexer comprising an indexer projection shaped to align the tote in a specified orientation; a set of collapser brackets coupled to the mounting frame, the set of collapser brackets comprising a first collapser bracket, a second collapser bracket, a third collapser bracket, and a fourth collapser bracket, the first and the second collapser brackets attached to the first frame member and spaced from each other, the third and the fourth collapser brackets attached to the second frame member and spaced from each other, each collapser bracket of the set of collapser brackets comprising: a bumper comprising a bumper projection shaped to engage with a latch on a sidewall of the tote; and a retainer comprising a retainer projection extending perpendicularly from the indexer and configured to engage with top side of the sidewalls; and a linear actuator configured to move the first frame member relative to the second frame member, wherein when the linear actuator is actuated the first frame member moves towards the second frame member to cause the set of collapser brackets to undergo motion toward the tote, wherein in response to the motion, the bumper engages with the latch and unlatches the latch on the sidewall of the tote to facilitate collapse of the sidewalls of the tote, and wherein the retainer keeps the sidewall from moving upward.
2. The tote collapser system of claim 1, wherein the indexer is formed as a part of each collapser bracket of the set of collapser brackets, wherein each collapser bracket comprises a receiving space between the bumper and the indexer to receive a thickness portion of the sidewall of the tote, wherein the indexer has an inclined surface angled away from the receiving space and the bumper, and wherein in response to the motion, the indexer aligns the tote relative to the mounting frame and guides the thickness portion of the sidewall of the tote toward receipt within the receiving

space.
3. The tote collapser system of claim 2, wherein the bumper has a contoured surface configured to progressively push the latch of the tote to unlatch the sidewall of the tote, wherein the contoured surface bounds a portion of the receiving space and curves away from the receiving space.
4. The tote collapser system of claim 3, wherein the bumper comprises a flat surface extending from the contoured surface, the flat surface configured to engage with a face of the sidewall and collapse the sidewall.
5. The tote collapser system of claim 1, wherein the indexer has a length greater than the bumper.
6. A tote collapser, comprising: a mounting frame comprising a frame member having a first end and an opposite second end; a first collapser bracket attached to the first end of the frame member; a second collapser bracket attached to the second end of the frame member; wherein each of the first and the second collapser bracket comprises: an indexer shaped to position a tote in a specified orientation; a bumper shaped to engage with a latch on a sidewall of the tote; and a receiving space between the indexer and the bumper to receive a thickness portion of the sidewall of the tote.
7. The tote collapser of claim 6, wherein the indexer has an inclined surface angled away from the receiving space and the bumper.
8. The tote collapser of claim 6, wherein the bumper has a contoured surface configured to progressively push the latch of the tote and unlatch the sidewall, wherein the contoured surface bounds a portion of the receiving space and curves away from the receiving space.
9. The tote collapser of claim 8, wherein the bumper comprises a flat surface extending from the contoured surface, the flat surface configured to engage with a face of the sidewall and collapse the sidewall.
10. The tote collapser of claim 6, wherein the frame member is fixed to a wall or support surface.
11. The tote collapser of claim 6, further comprising: a second frame member movable with respect to the frame member.
12. The tote collapser of claim 11, wherein the frame member and the second frame member are coupled to a linear actuator configured to automatically move the frame member and the second frame member upon actuation.
13. The tote collapser of claim 11, wherein the frame member and the second frame member are coupled to lever configured to manually move the frame member and the second frame member.
14. The tote collapser of claim 6, further comprising a third collapser bracket attachable to a hand glove, the third collapser bracket comprising an indexer, a bumper, and a receiving space.
15. A method of collapsing sidewalls of a tote using a tote collapser, comprising: receiving a tote at a mounting frame of the tote collapser, the mounting frame comprising a first frame member and a second frame member relatively movable with respect to each other; actuating a linear actuator to move a first frame member relative to a second frame member of the mounting frame; and moving, via the linear actuator, a set of collapser brackets coupled to the first frame member and the second frame member of the mounting frame to cause the set of collapser brackets to engage with and unlatch latches on the sidewalls of the tote and collapse the sidewalls of the tote, wherein the set of collapser brackets comprises: a first collapser bracket, a second collapser bracket, a third collapser bracket, and a fourth collapser bracket, the first and the second collapser brackets attached to the first frame member and spaced from each other, the third and the fourth collapser brackets attached to the second frame member and spaced from each other, and wherein each collapser bracket of the set of collapser brackets comprises: an indexer; a bumper; and a receiving space between the indexer and the bumper.
16. The method of claim 15, wherein moving the set of collapser brackets comprises: aligning, via the indexers in the set of collapser brackets, the tote in a specified orientation such that the first frame member is parallel to a first sidewall of the tote and the second frame member is parallel to a second sidewall of the tote, the second sidewall being opposite to the first sidewall.
17. The method of claim 16, wherein moving the set of collapser brackets comprises: subsequent to

the aligning, moving the set of collapser brackets to cause the bumper to engage with latches on the sidewalls of the tote; and collapsing, via the bumper, the sidewalls of the tote.

18. The method of claim 16, wherein the collapsing comprises: receiving portions of the sidewalls of the tote within respective receiving spaces between the indexer and the bumper.

19. The method of claim 16, wherein the linear actuator is a hydraulic piston configured to actuate automatically.

20. The method of claim 16, wherein the linear actuator comprises a rack and pinion coupled to a lever to manually actuate the linear actuator.